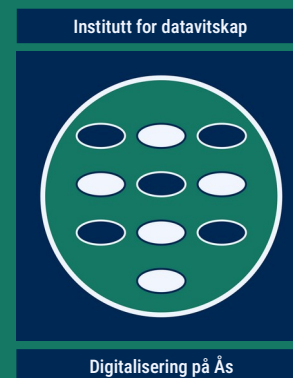
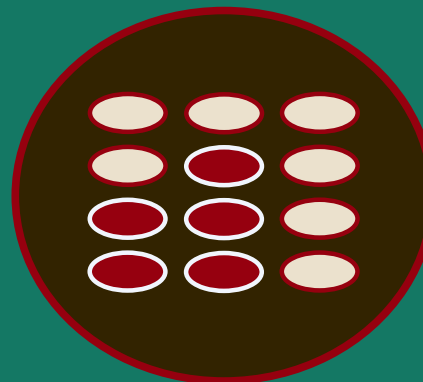


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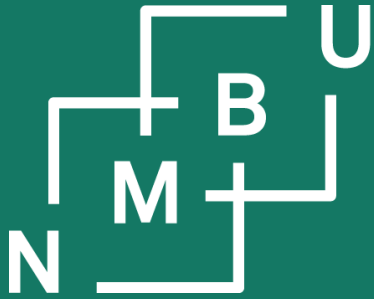
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Datahandtering og -analyse

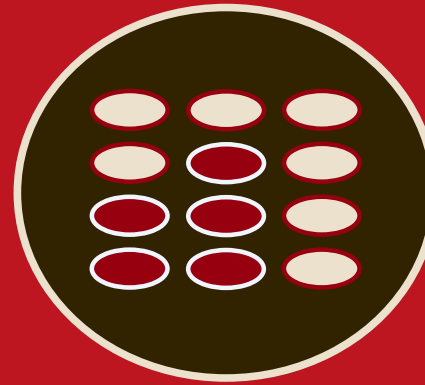
1 Ontologiar

- 1.1 Den semantiske veven
- 1.2 Vevontologispråket OWL
- 1.3 Terminologiutvikling

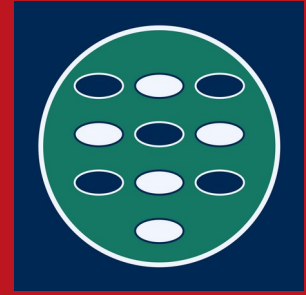
- 1.4 Ontologiutvikling
- 1.5 **BFO, DOLCE og EMMO**
- 1.6 **Semantisk heterogenitet**



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Digitalisering på Ås

1 Ontologiar

1.5 Grunnleggjande ontologiar

1.6 *Semantisk heterogenitet*

Termliste

nn **endurantisme** *m.*

en **endurantism**

nn **grunnleggjande ontologi** *m.*

en **foundational ontology**

nn **inngåande** *n.*

en **endurant**

nn **kvalitet** *n.*

en **quality**

nn **kvalitetsverdi** *n.*

en **quale**

nn **mereologi** *m.*

en **mereology**

nn **pågåande** *n.*

en **perdurant**

nn **perdurantisme** *m.*

en **perdurantism**

nn **representamen** *n.*

en **representamen**

Grunnleggjande ontologiar

6.1.1 Typical content of a foundational ontology

Foundational ontologies provide a high-level categorisation about the kinds of things that will be represented in the ontology, such as *process* and *physical object*, relations that are useful across subject domains, such as *participation* and *parthood*, and (what are and) how to represent ‘attributes’ in a particular subject domain, such as **Colour** and **Height** (recall Section 1.3), which can be done, e.g., as *quality* or some kind of *dependent continuant* or *trope*.

- One can commit to a 3-Dimensional view of the world with objects persisting in time or take a(4-Dimensional (perdurantist) stance with space-time worms; e.g., are you convinced that you after reading this sentence is a different you than you before reading this sentence? If so, then you may well be a perdurantist, if you consider yourself to be the very same entity before and after, then you lean toward the 3D, endurantist, commitment (but before proclaiming to be one or the other based on this single example, do read up on the details and the implications).

There are more of such questions and decision to make. If you do not want to entertain yourself with these questions, you can take someone else’s design decisions and use that in ontology development. Someone else’s design decisions on Ontology for a set of such questions typically is available in a **foundational ontology**, and the different answers to such questions end up as different foundational ontologies.

Grunnleggjande ontologiar

6.1.2 Several foundational ontologies

In this section, a selection of the foundation ontologies are summarised with respect to their ontological commitments.

Descriptive Ontology for Linguistic and Cognitive Engineering (DOLCE)

As the name suggests, DOLCE has a strong cognitive/linguistic bias. It takes a descriptive attitude and the categories mirror cognition, common sense, and the lexical structure of natural language. The emphasis is on cognitive invariants and the categories are intended as ‘conceptual containers’ in the sense that there are no deep metaphysical implications. Its documentation [203] focuses on the design rationale to facilitate a comparison with different ontological options.

It is rigorous, systematic, and has a rich axiomatisation.

Basic Formal Ontology (BFO)

The BFO developers see Ontology as reality representation. It aims at reconciling the 3-dimensionalist and 4-dimensionalist views

and is heavily influenced by parthood relations, boundaries, and dependence. Around the same time, also a BFO Core was released (in FOL, Isabelle syntax), where BFO entities were built up from mereology.

Grunnleggjande ontologiar

Table 6.1: Common alignments between DOLCE-Lite, BFO and GFO; the ones numbered in bold can also be mapped. (Source: [KK13a])

	DOLCE-Lite	BFORO	GFO
Class			
1.	endurant	Independent Continuant	Presential
2.	physical-object	Object	Material_object
3.	perdurant	Occurrent	Occurrent
4.	process	Process	Process
5.	quality	Quality	Property
6.	space-region	SpatialRegion	Spatial_region
7.	temporal-region	Temporal-Region	Temporal_region

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Chapter 6. Top-down Ontology Development

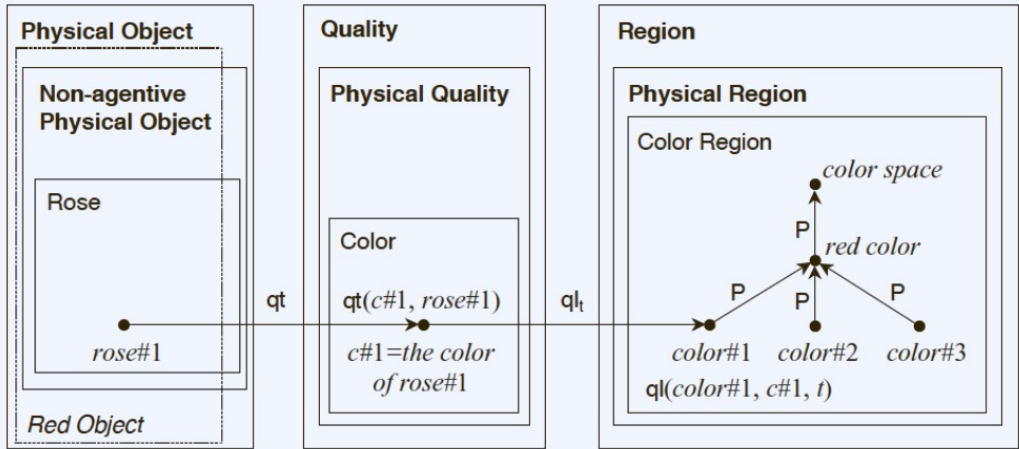
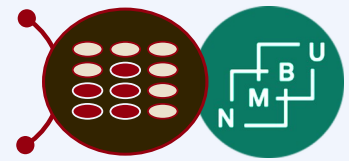


Figure 6.2: DOLCE's approach for qualities ('attributes') (Source: [MBG⁺03])



Mereologi, ontologi og metaontologi

Grunnleggjande ontologiar formaliserer bl.a. korleis ting kan ha eigenskapar. Til dømes er det i DOLCE sånn:

- dol:**quale** (**kvalitetsverdi**): mogleg verdi på ein kvalitet
eksempel: "raudt" er ein kvalitetsverdi av di at noko kan ha raudt som farge, dvs. at "raudt" er ein mogleg verdi for farga.
- dol:**quality** (**kvalitet**): eigenskap (ved eit objekt) som har ein kvalitetsverdi
eksempel: farga til ein blome; kvalitetsverdien kan forandre seg over tid.

120

Chapter 6. Top-down Ontology Development

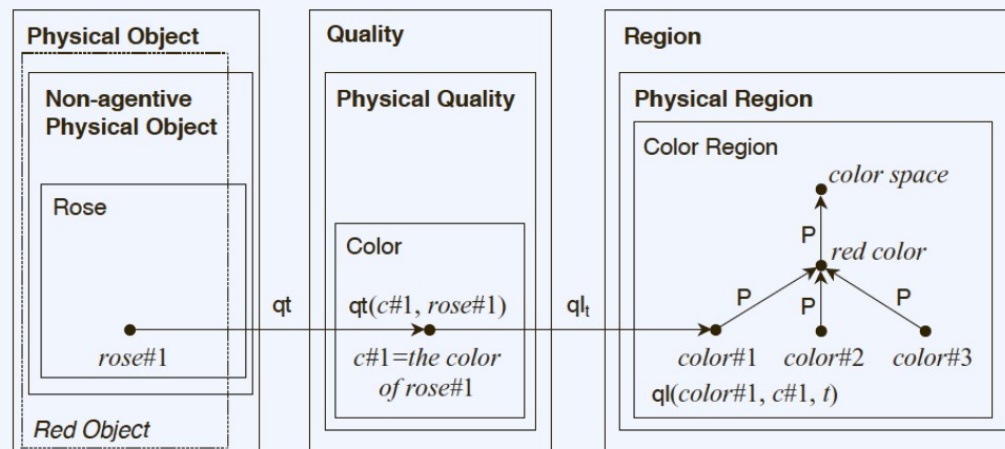
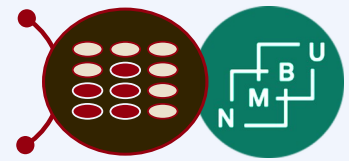


Figure 6.2: DOLCE's approach for qualities ('attributes') (Source: [MBG⁺03])



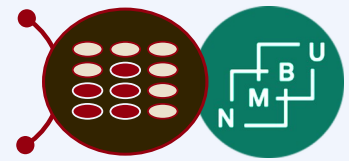
Mereologi, ontologi og metaontologi¹

Nokre andre utvalde spørsmål og skiljelinjer om «kva som finst»: ^{1, 2}

- *Nominalisme* eller *Platonisme* (kallast òg for realisme)
- Behandling av teikn, t.d. *semiotikk etter Saussure* eller *etter Peirce*
- *Eksisterer hypotetiske ting*, moglege og til og med umoglege? Om ja, korleis og kvar eksisterer dei? (Ekstremtilfelle: «Meinongs jungel» ...) Om nei, korleis kan vi seia noko om dem? (Tenk: Anselms gudsbevis)
- Korleis handterer vi det at *forholda kan vera tidsavhengige*; er det *noko som uansett blir uforandra*, t.d. identiteten av objekt eller eigenskaper?

¹F. Berto, M. Plebani, *Ontology and Metaontology*, Bloomsbury, **2015**.

²M. Horsch, S. Chiacchiera, *et al.*, «Semantic interoperability based on the European Materials and Modelling Ontology and its ontological paradigm: Mereosemiotics». In *Proc. WCCM-ECCOMAS Congress 2020*, doi:10.23967/wccm-eccomas.2020.297, **2021**.



Endurantism and the ontology BFO

nn **pågåande** *n.*; en **perdurant** (i BFO: «**occurrent**»)

prosess eller noko som liknar ein prosess, og som – i tillegg til romlege delar – også har tidslege delar

- Eksempel: Eit kurs består av fleire førelesingar, øvingar, ..., og eit eksamen. Desse er alt pågåande.

nn **inngåande** *n.*; en **endurant** (i BFO: «**continuant**»)

noko som beheld sin identitet over tid, og som kan ha romlege delar, men ikkje tidslege delar

- Eksempel: Ein person kan gå gjennom barndommen fyrst, så gjennom vaksenalderen. Vi kan uansett omtala til det som eitt individ: «Johannes var barn, og så er han vaksen (på eit anna tidspunkt).» Begge gongar handlar det om Johannes.
- Eller: Under førelesing og under eksamen er det den same studenten. Den har ikkje ein tidsdel «førelesingsstudent» og ein annan som er «eksamensstudent».
- **Paradigmet** der omgrepa «*inngåande*» og «*pågåande*» blir brukt, **er endurantismen**. Den står *i motsetning til perdurantismen*, som hevdar at det berre finst pågåande ting, som både har romlege og tidslege delar.

Endurantism and the ontology BFO^{1, 2}

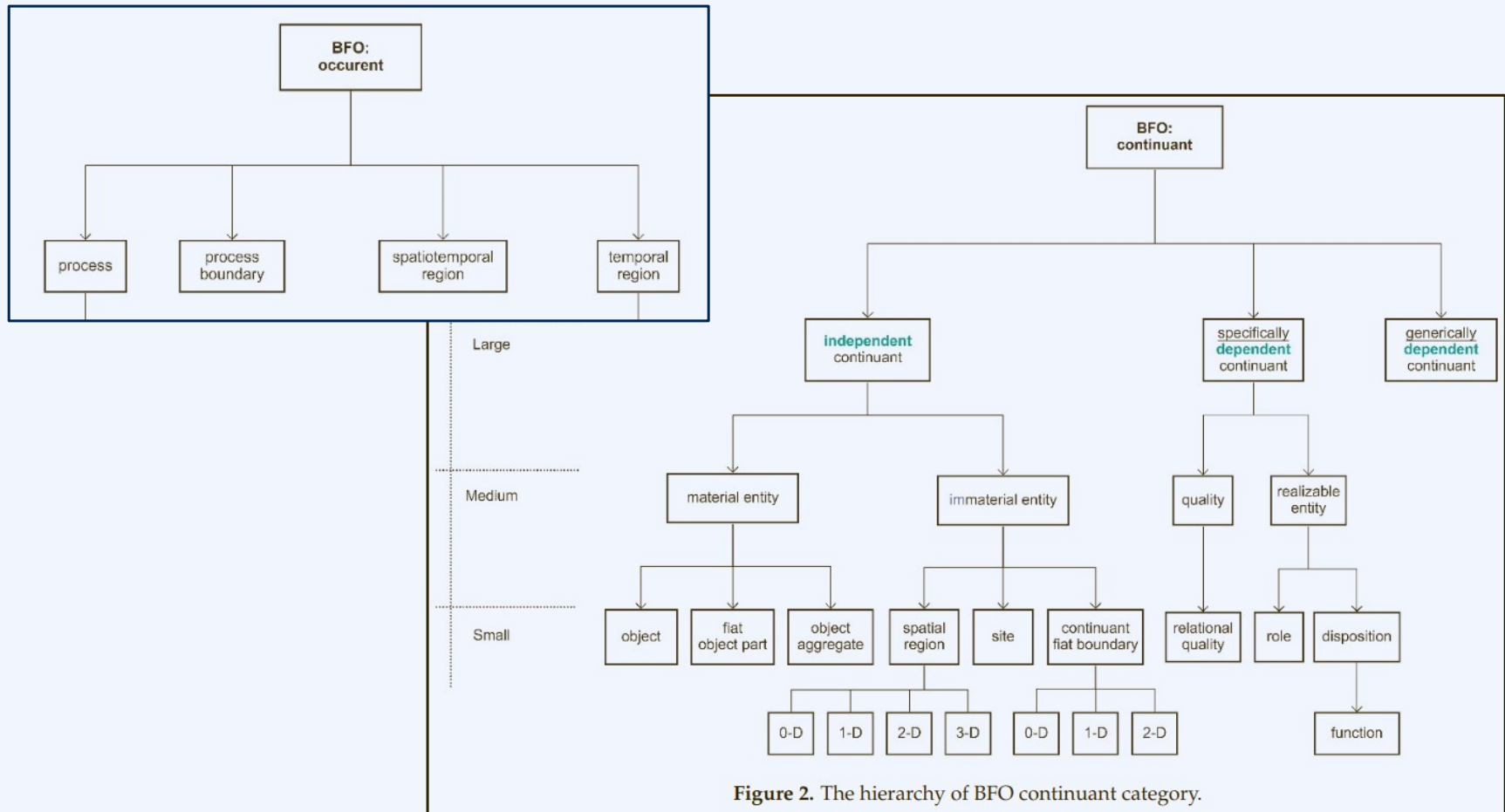


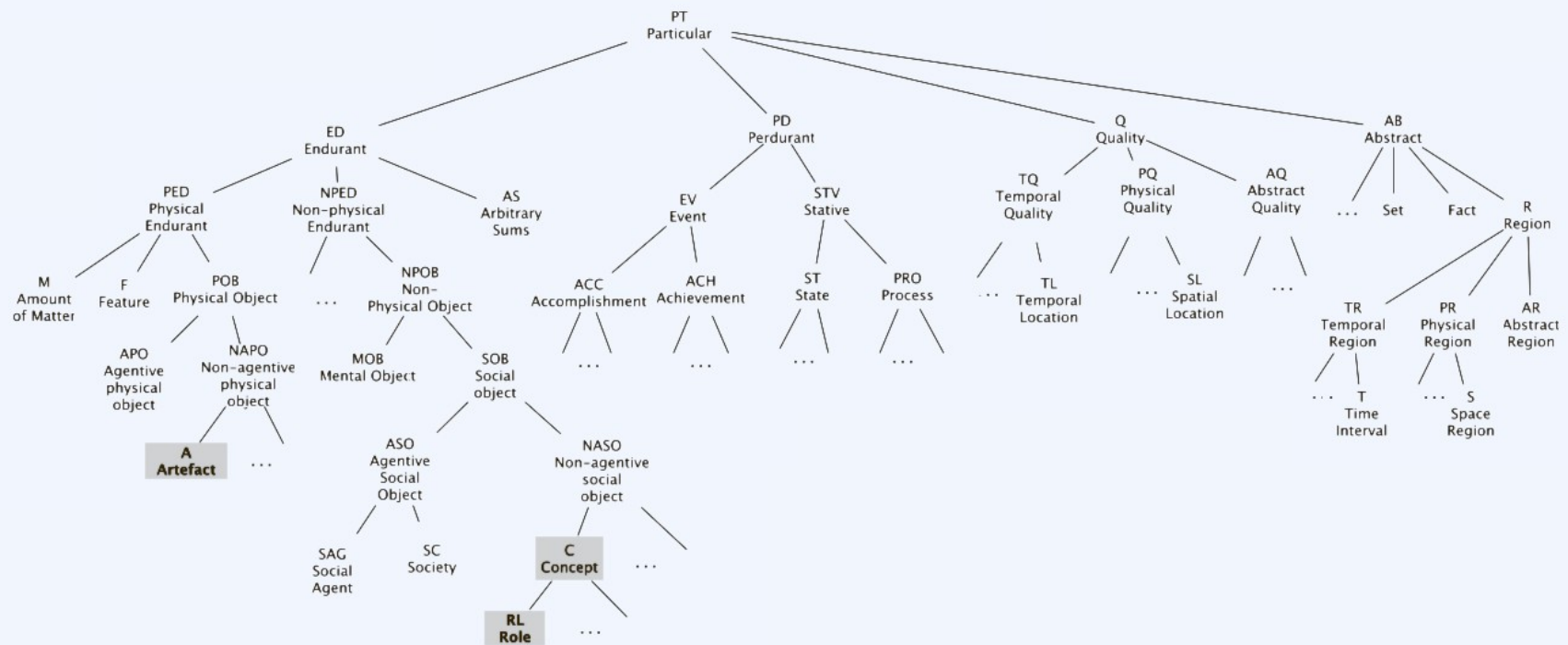
Figure 2. The hierarchy of BFO continuant category.

¹R. Arp, B. Smith, A.D. Spear, *Building Ontologies with Basic Formal Ontology*, Cambridge, MA: MIT Press (ISBN 978-0-262-52781-1), **2015**

²C. Pietra, R. De Lotto, R. Bahshwan, *Sustainability* **13**(23): 13844, **2021**

Ontologien DOLCE¹

As depicted in the taxonomy in Figure 1, the basic categories of DOLCE are endurant (aka continuant), perdurant (occurrent), quality, and abstract.



¹S. Borgo, R. Ferrario, et al., «DOLCE: A descriptive ontology for linguistic and cognitive engineering», *Applied Ontology* **17**: 45-69, doi:10.3233/ao-210259, **2022**

Ontologien DOLCE¹

The relation connecting endurants and perdurants is called *participation*. An endurant can *be* in time by participating in a perdurant, and perdurants *happen* in time by having endurants as participants. For instance, a person is in time by participating to her own life, and a conference talk happens if at least one presenter (or attendant) participates to it.

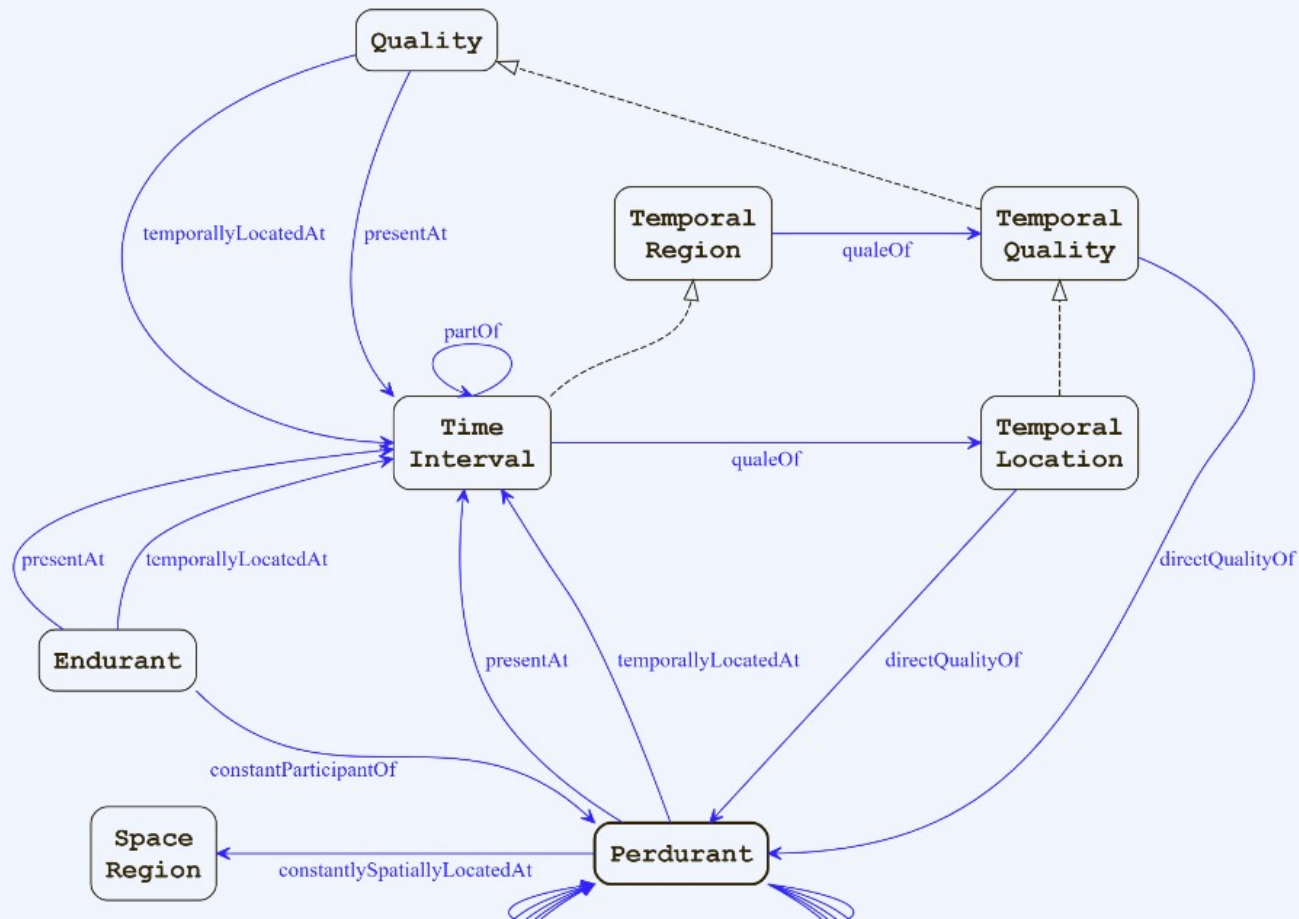
II. Independent vs. dependent entity.. This distinction is found across the entire taxonomy of DOLCE. For instance, features (e.g., edges, holes, bumps, etc.) are endurants whose existence depends on some physical object (the feature bearer), while physical objects are independent entities, i.e., their existence does not require other endurants to exist.

III. Processes vs. events.. In DOLCE processes and events are special types of perdurants. As it can be seen from Figure 1, DOLCE covers various classes of perdurant following taxonomic distinctions

Common examples of stative perdurants are states; e.g., a sitting state is stative because the sum of two sittings is still a sitting. Among stative perdurants, processes are cumulative but not homeomeric, namely, they have parts of different types; e.g., there are (even very short) temporal parts of a running that are not themselves runnings. Finally, eventive occurrences (events) are not cumulative, and they are called achievements if they are atomic, otherwise they are accomplishments.⁵

¹S. Borgo, R. Ferrario, et al., «DOLCE: A descriptive ontology for linguistic and cognitive engineering», *Applied Ontology* **17**: 45-69, doi:10.3233/ao-210259, **2022**

Ontologien DOLCE^{1,2}

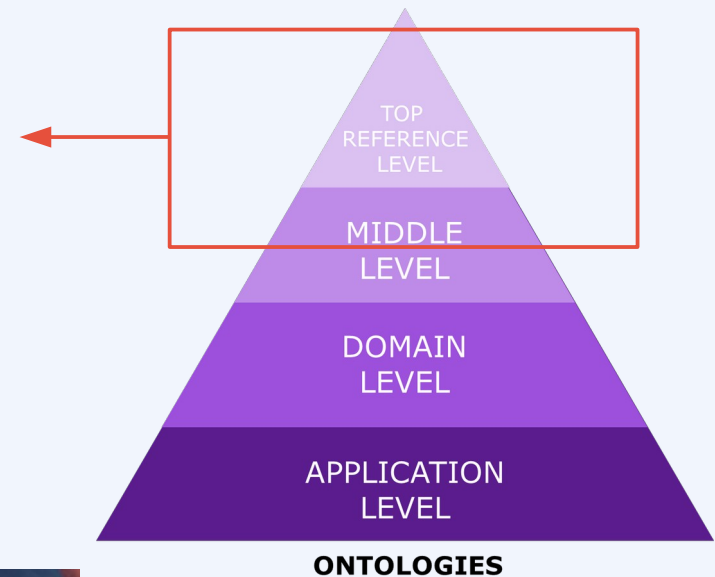


¹<https://github.com/appliedontolab/DOLCE/blob/main/OWL/DOLCEbasic.owl>

²D. Porello, L. Vieu, et al., «DOLCE in OWL: The core theory», in *Proc. JOWO 2024*, 2024

European initiatives

- **Community development** that brought together applied ontology, industrial digitalization, and computational science and engineering.
- OntoCommons ecosystem¹ (OCES) including the three **foundational ontologies** BFO, DOLCE, and EMMO. Development of **bridge concepts**² as a technique for ontology alignment.



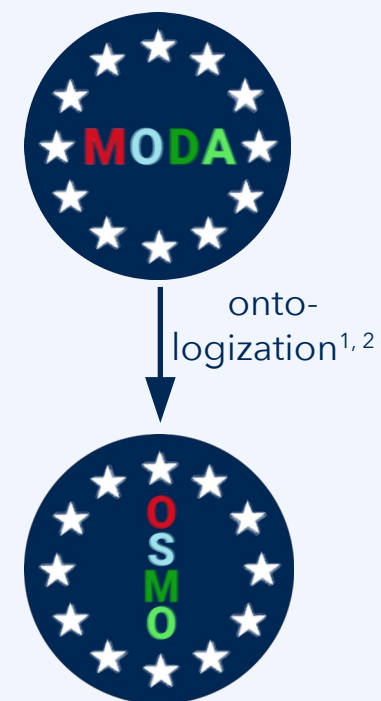
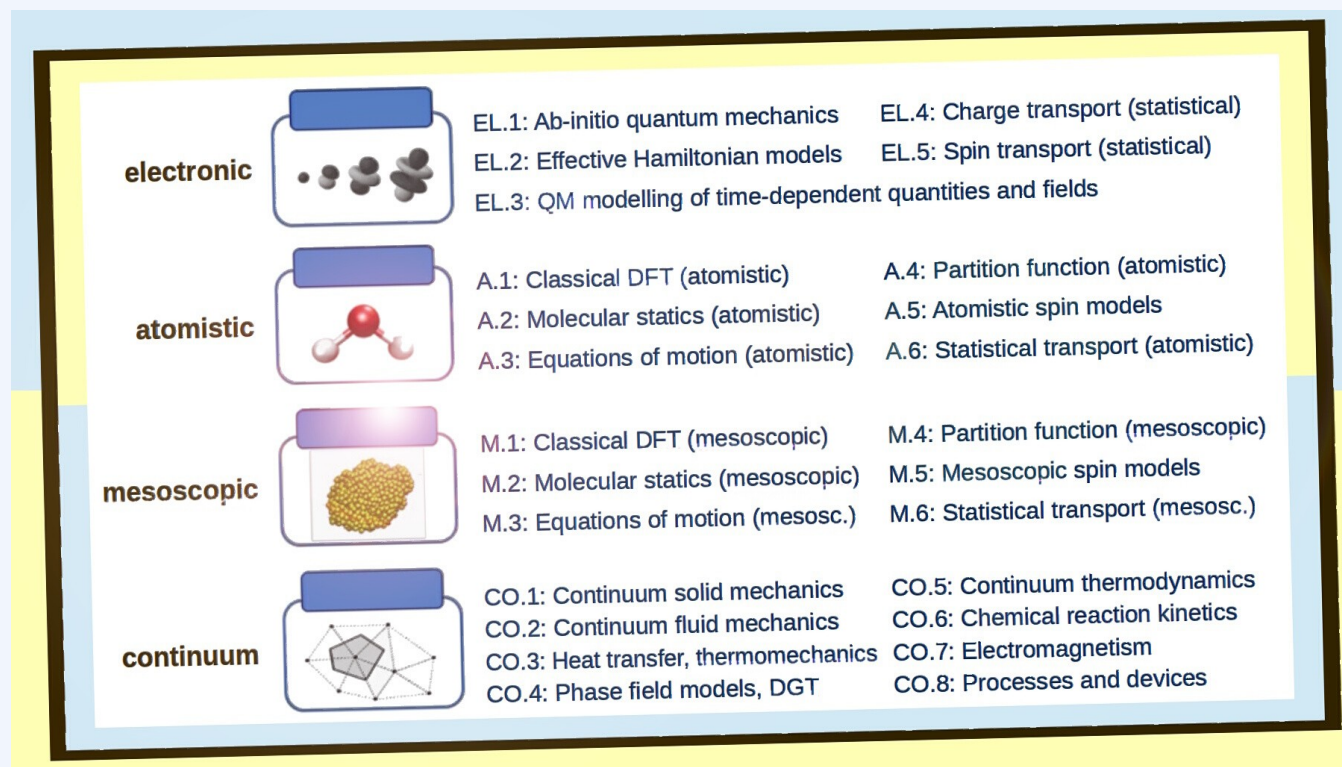
¹M. Magas, D. Kiritsis, *Int. J. Production Res.* **60**(2): 479–492, doi:10.1080/00207543.2021.1989514, **2022**.

²A. de Baas et al., *IEEE Access* **11**: 120372–120401, doi:10.1109/access.2023.3327725, **2023**.



European initiatives

As an attempt at metadata standardization, RoMM/MODA resulted in a closed epistemic space with a rigid categorization of modelling methodologies. MODA/CHADA documentations are hard to create and **hard to use by humans, but not machine-actionable**.

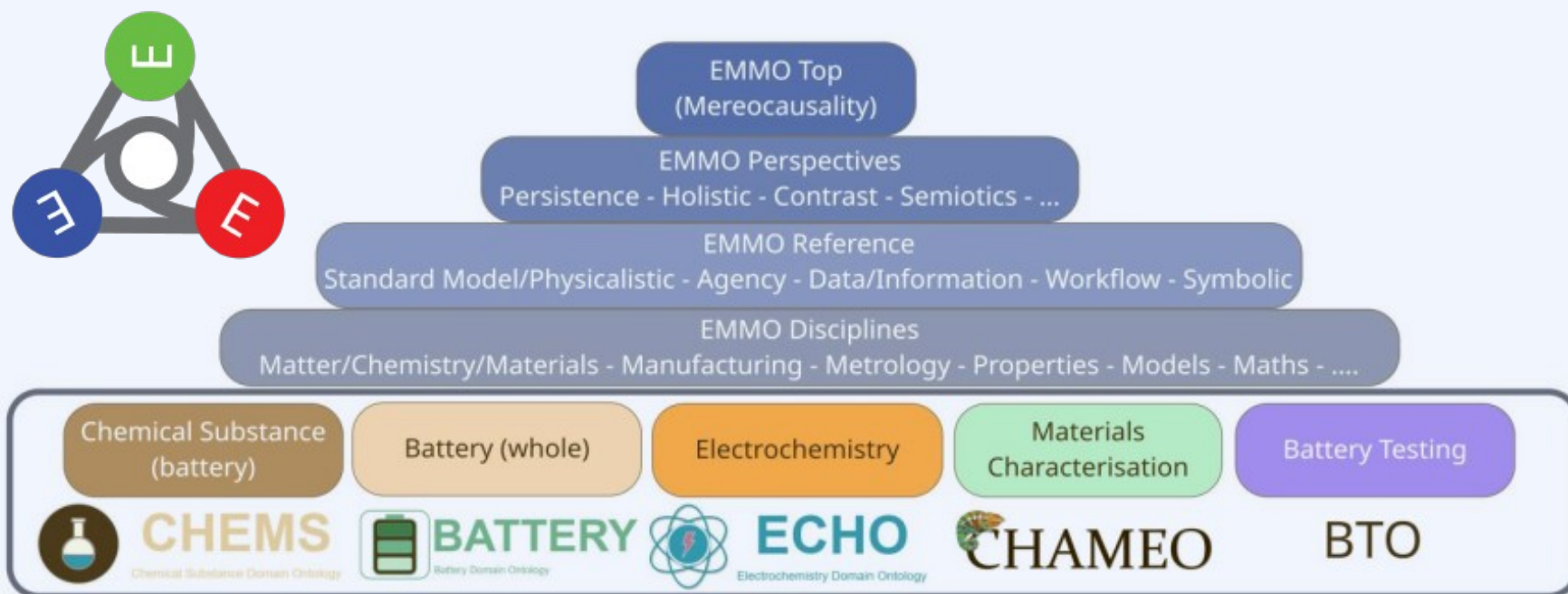


¹M. T. Horsch *et al.*, *J. Chem. Eng. Data* **65**(3): 1313–1329, doi:10.1021/acs.jced.9b00739, **2020**.

²M. T. Horsch *et al.*, in *Proc. JOWO 2021*, CEUR vol. **2969**: p. 47 (FOIS ontology showcase), **2021**. 15

Perdurantism and the ontology EMMO¹

EMMO: Elementary Multiperspective Material Ontology



«EMMO builds upon several theoretical frameworks: Semiotics. Mereotopology. Physics. Metrology. Description logic.»

¹<https://emmo-repo.github.io/versions/1.0.0-beta/emmo.pdf>

Perdurantism and the ontology EMMO¹

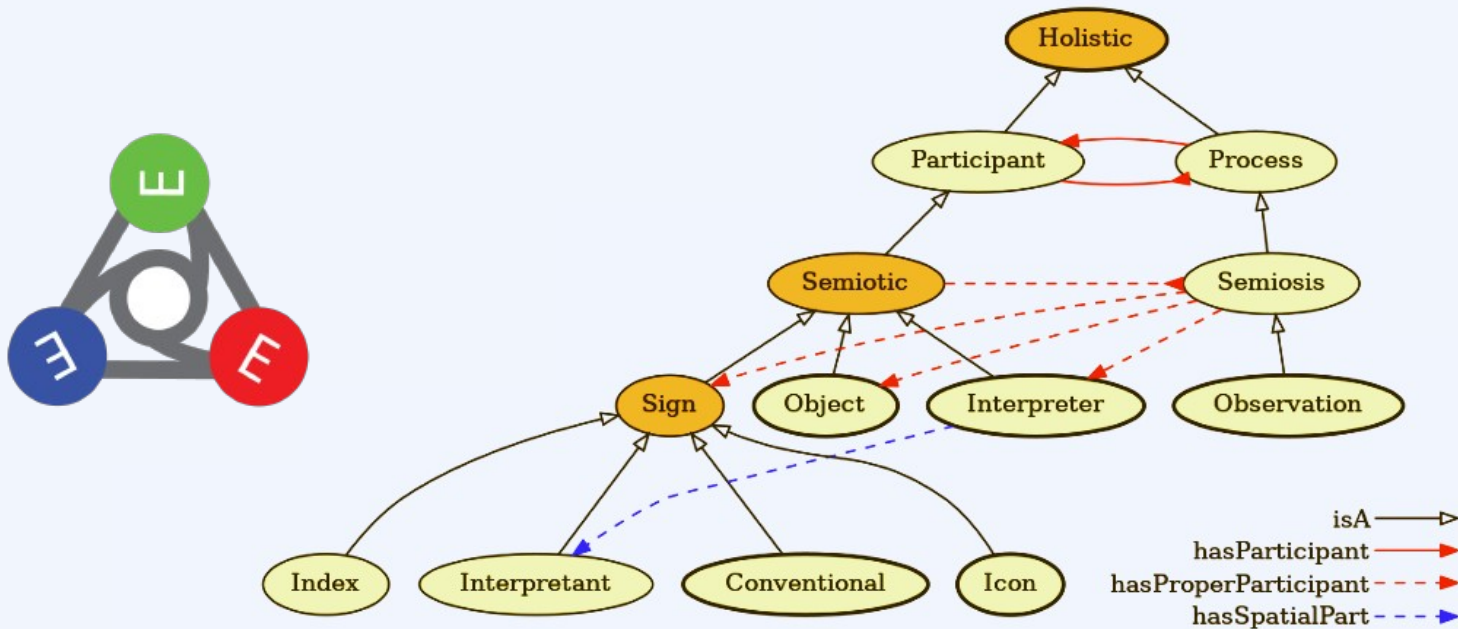


Figure 1.9: The semiotic level, showing both the taxonomy (open black arrows) and other relations as listed in the caption. The inverted arrows corresponds to inverse relations.

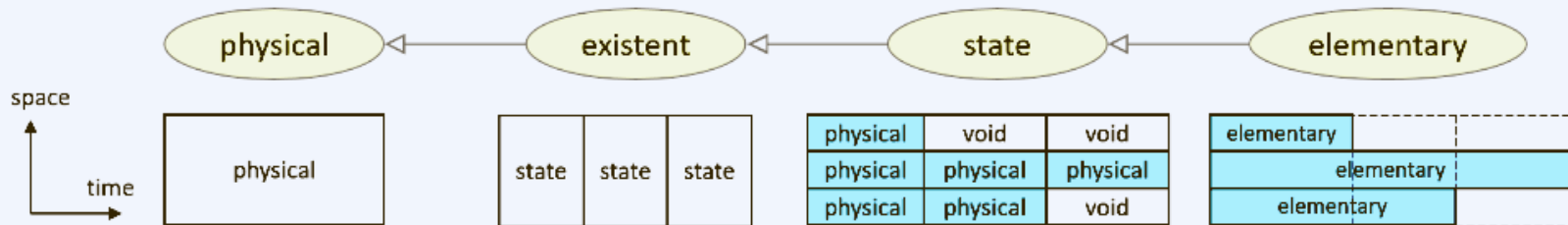


Figure 1.6: Parthood hierarchy under Physical.

¹<https://emmo-repo.github.io/versions/1.0.0-beta/emmo.pdf>

Discussion: Participants and processes

In endurantism, there are processes (perdurants) and participants (endurants).



How does this look for: Napoleon was a soldier.

Later, Napoleon was an emperor.

Napoleon went to war (participated in process of a war).

Later, he was exiled (underwent a state of exile).

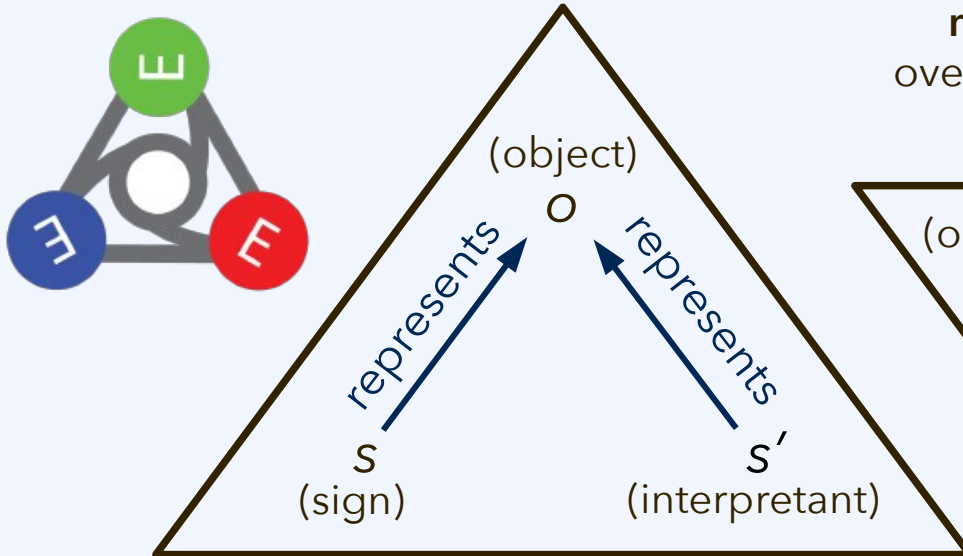
In perdurantism, there are no such two distinct categories. Endurantists say that «in perdurantism, there are only perdurants», but this means using the words from endurantism. Perdurantism does not use these categories at all.

How does the above look in perdurantism / four-dimensionalism?

- What form does participation take?
- How is it related to mereology?
- How is it more flexible in that the roles of being a process and being a participant are not disjoint? The same object can be both ...

Semiotikk

Peirces semiotiske trekant: **Teikn** (1.), objekt (2.) og nytt teikn: **Interpretant** (3.).



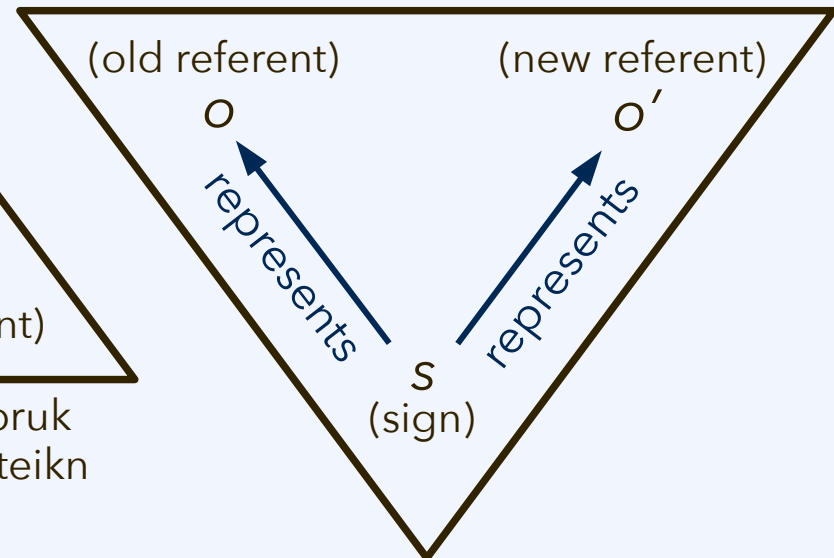
semiose, dvs. prosess av teiknbruk som fører til danning av eit nytt teikn



C. S. Peirce, "Logic as semiotic: The theory of signs," **1897, 1903**.
In: Buchler, utg., *Philosophical Writings of Peirce*, Dover, **1955**.

C. S. Peirce (1839–1914)

metonymisering, dvs. prosess av overføring av teiknet til eit anna objekt (seinare utvikling i semiotikk)

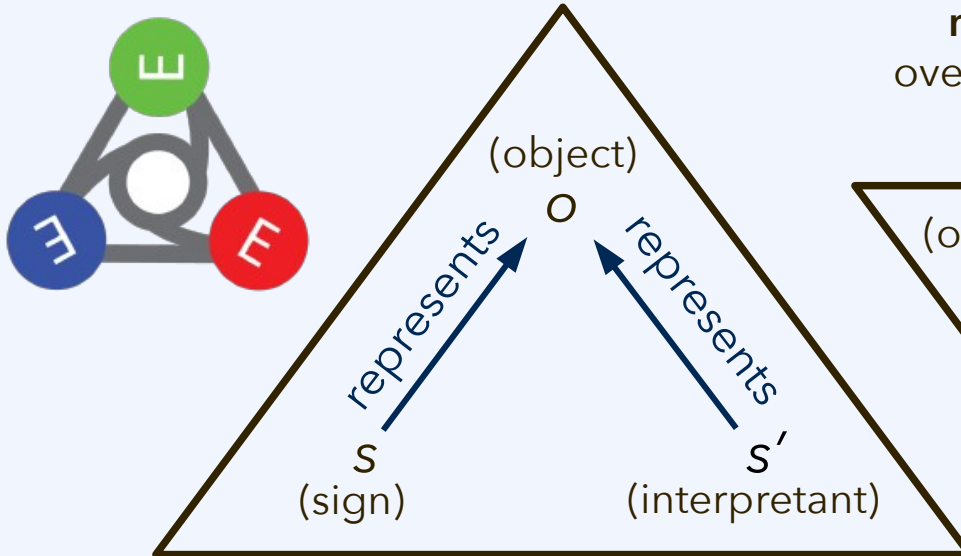


Jf. òg P. Klein *et al.*, in *Proc. JOWO 2021*, CEUR-WS, <http://ceur-ws.org/Vol-2969/paper26-FOMl.pdf>, **2021**.

Semiotikk

Peirce kallar teikn (inkl. interpretantar) òg for **representamina**.

Peirces semiotiske trekant: **Teikn** (1.), objekt (2.) og nytt teikn: **Interpretant** (3.).



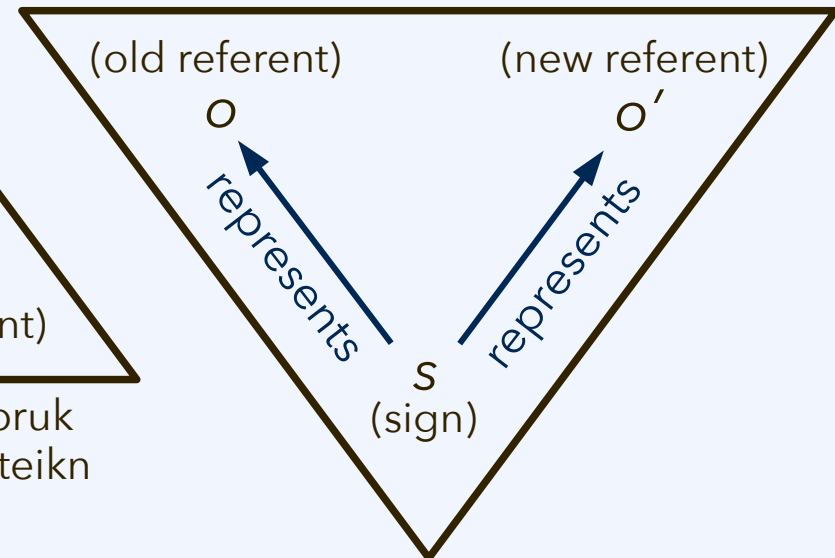
semiose, dvs. prosess av teiknbruk som fører til danning av eit nytt teikn



C. S. Peirce, "Logic as semiotic: The theory of signs," **1897, 1903**.
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C. S. Peirce (1839–1914)

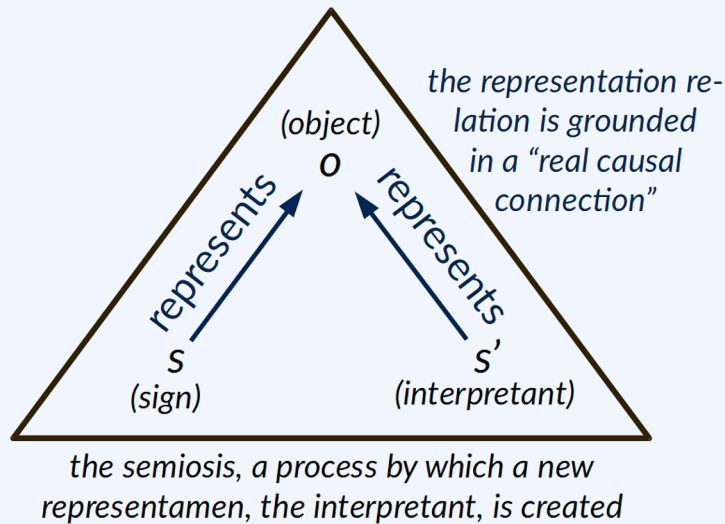
metonymisering, dvs. prosess av overføring av teiknet til eit anna objekt (seinare utvikling i semiotikk)



Jf. òg P. Klein *et al.*, in *Proc. JOWO 2021*, CEUR-WS, <http://ceur-ws.org/Vol-2969/paper26-FOMl.pdf>, **2021**.

Research workflows as cognitive processes

Peircean semiotics



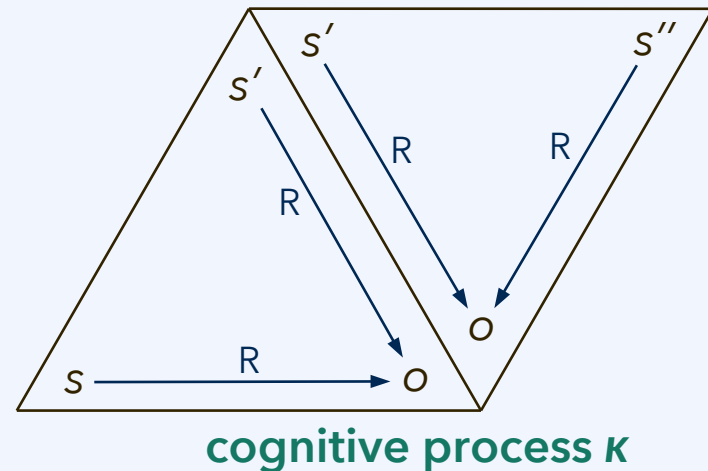
Each cognitive step starts from one representation relation, e.g., R_{so} , and creates a new one, $R_{s'o}$.

The successor step reuses $R_{s'o}$ and creates the next relation, $R_{s''o}$.

Cognitive process (example):

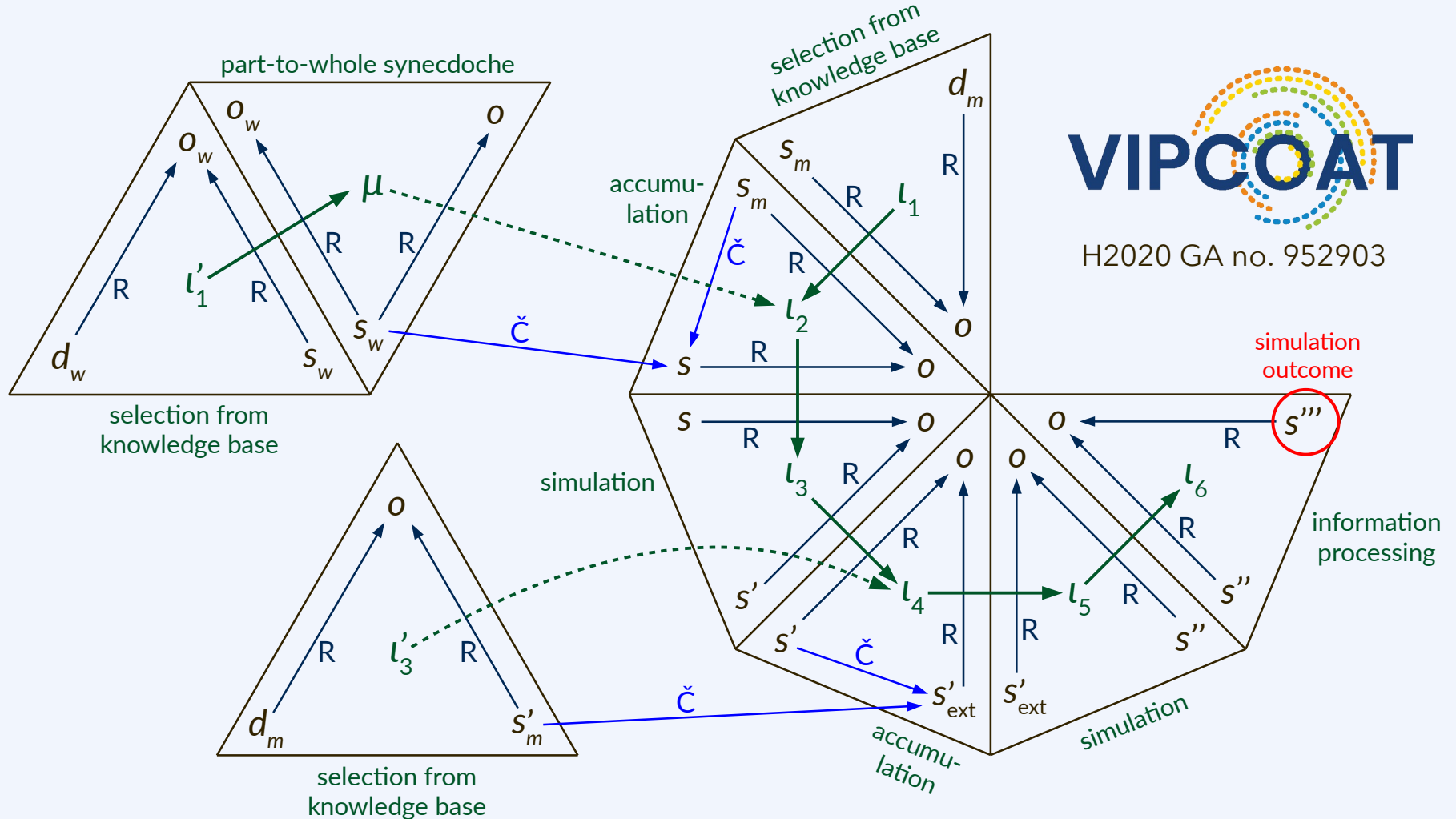
- First, experimental data s for material o are used to parameterize a model, obtaining model s' .
- Then, a simulation is done using model s' , yielding the simulation result s'' (which also represents o).

Research workflows as cognitive processes:¹



¹In *Proc. JOWO 2021*, CEUR vol. **2969**: p. 3 (FOUST), **2021**.

Research workflows as cognitive processes¹

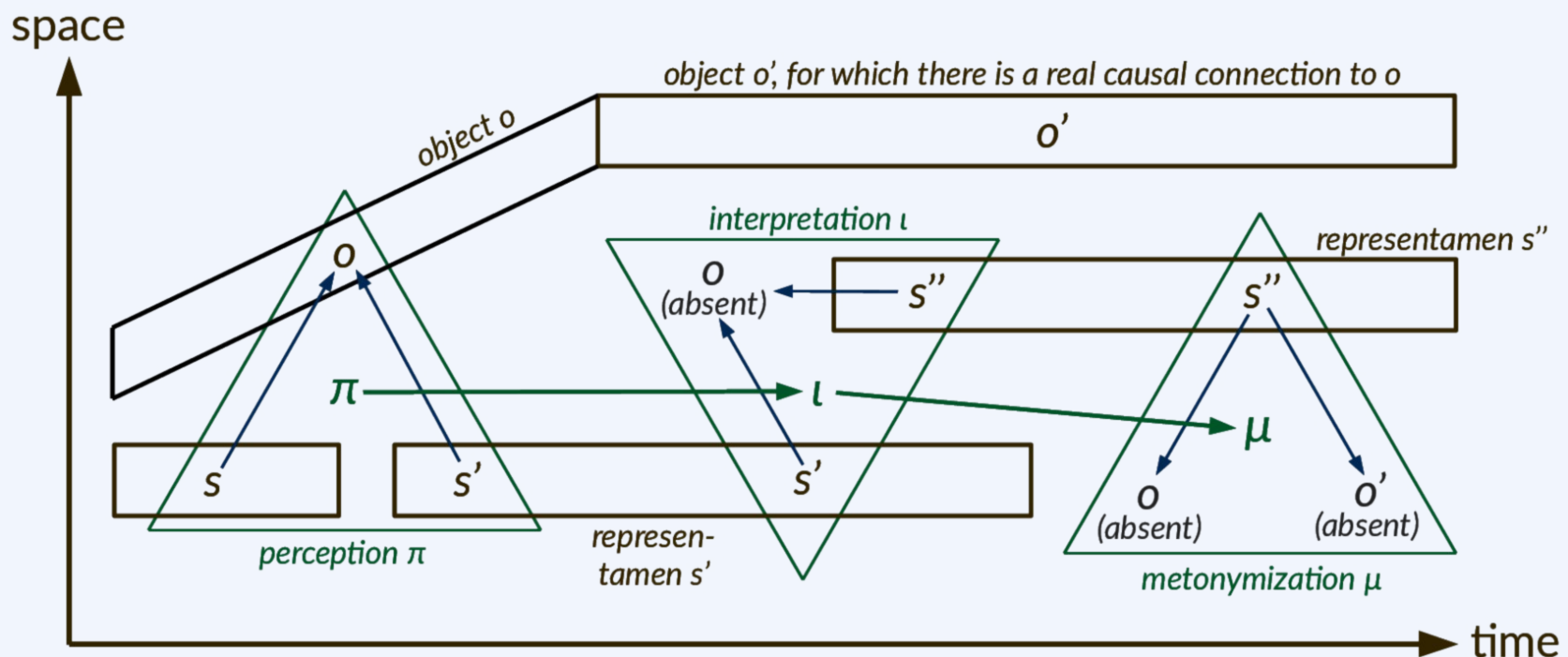


¹P. Klein et al., no. 26 in *Proc. JOWO 2021*, 2021.

4D physicalism and semiotics

PIMS-II mid-level ontology:^{1,2} <http://www.molmod.info/semantics/pims-ii.ttl>

Mereosemiotics:¹⁻³ Combination of mereotopology and Peircean semiotics



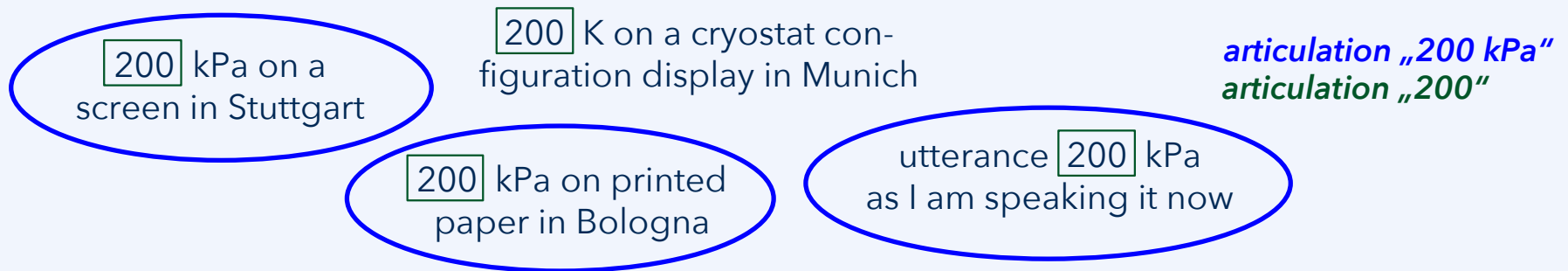
¹M. T. Horsch, no. 3 in *Proc. JOWO 2021*, **2021**.

²P. Klein et al., no. 26 in *Proc. JOWO 2021*, **2021**.

³M. T. Horsch, S. Chiacchiera, B. Schembera, M. Seaton, I. T. Todorov, in *Proc. ECCOMAS 2020*, **2021**.

4D physicalism and semiotics

„200 kPa“ is **one well-defined value**, it needs to be capable of **acting collectively** (not just each instance) as representational element.¹ However, EMMO is based on nominalism with objects defined by 4D spacetime regions.



Another example: „The stadiums of the Premier League have a total capacity of 833 000 people.“ The referent of the property are all the stadiums together, as a collective; none of the individual stadiums holds 833 000 people.

EMMO defines items (contiguous regions) and mereotopological collectives („Collections“); in PIMS-II, these are supplemented by **semiotic collectives**.

¹M. Horsch, S. Chiacchiera, B. Schembera, M. Seaton, I. Todorov, *Proc. WCCM-ECCOMAS 2020*, **2021**.

Glossar

grunnleggjande ontologi

Definisjon: ontologi som spesifiserer «hva som egentlig mest grunnleggende sett faktisk finnes. [...]» (E. D. Bøhn i SNL om metafysikk)

- «Finnes det for eksempel egentlig tall?» (ibid.)
- I anvend ontologi er det vanleg praksis å skilja mellom **domene-**, **mellomnivå-** og **grunnleggjande** ontologiar.
- Viktige grunnleggjande ontologiar er DOLCE og BFO, som representerer endurantismen, og EMMO, som representerer perdurantismen.

mereologi

Definisjon: delområde innanfor ontologi og anvend logikk som behandlar forhold mellom delar og heilskap

Glossar

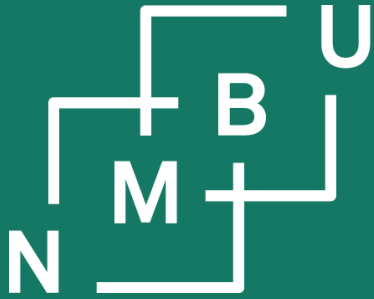
grunnleggjande ontologi // foundational ontology

Definisjon: ontologi som spesifiserer «hva som egentlig mest grunnleggende sett faktisk finnes. [...]» (E. D. Bøhn i SNL om metafysikk)

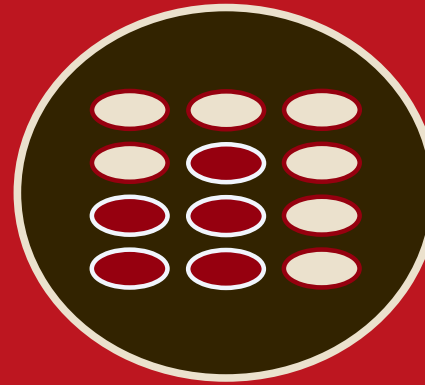
- «Finnes det for eksempel egentlig tall?» (ibid.)
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- Viktige grunnleggjande ontologiar er DOLCE og BFO, som representerer endurantismen, og EMMO, som representerer perdurantismen.

mereologi // mereology

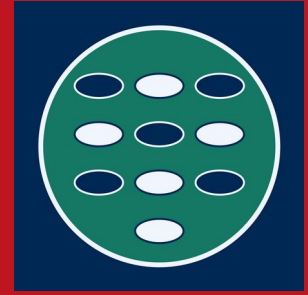
Definisjon: delområde innanfor ontologi og anvend logikk som behandlar forhold mellom delar og heilskap



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Institutt for datavitenskap

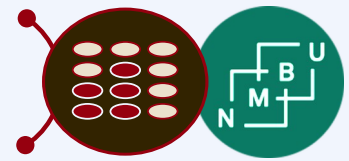


Digitalisering på Ås

1 Ontologiar

1.5 Grunnleggjande ontologiar

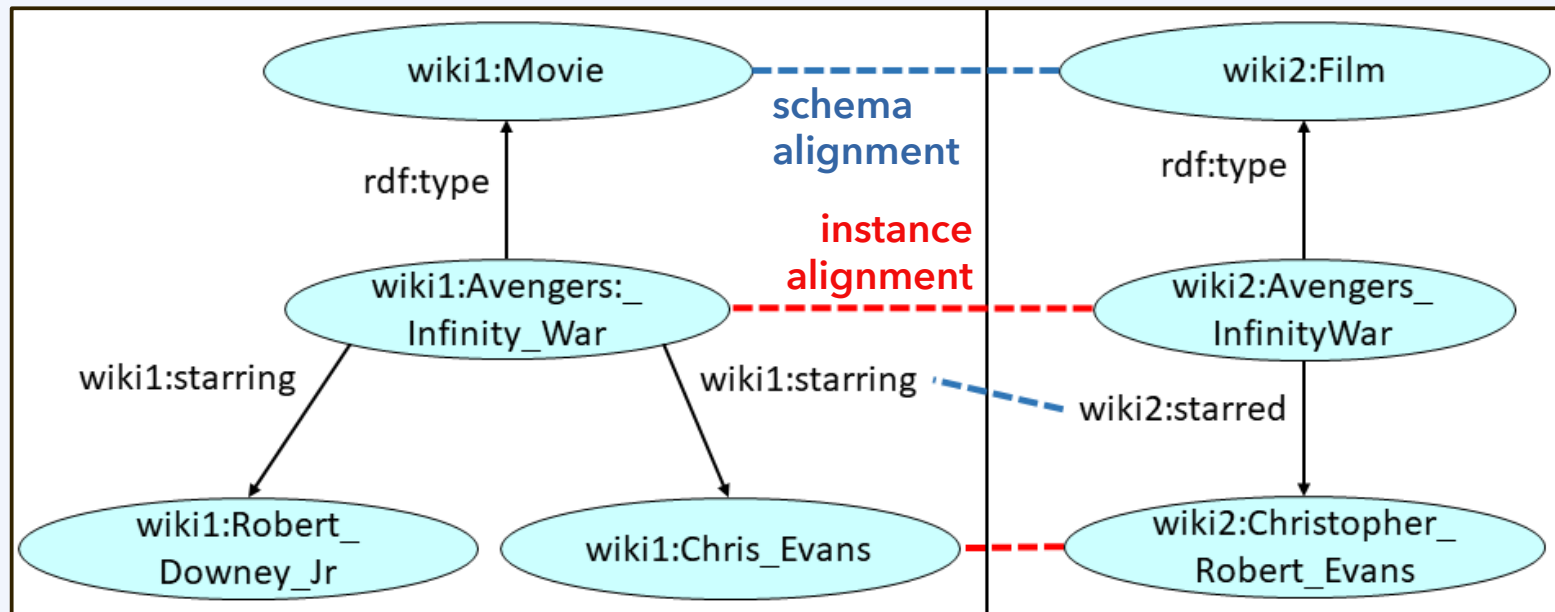
1.6 Semantisk heterogenitet



Semantic heterogeneity

Different platforms and tools use different metadata standards (ontologies).

Example from the Ontology Alignment Evaluation Initiative:¹



In many cases, the knowledge graph structure required by source and target schemas/ontologies can be very different, so the mapping is hard to specify.

¹<https://oaei.ontologymatching.org/2025/knowledgegraph/>

Use of foundational ontologies for alignment

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C. Trojahn et al. / Foundational ontologies meet ontology matching

Table 2
Summary of matchings of foundational ontologies (*<http://www.thezfiles.co.za/ROMULUS/ontologyAlignment.html>)

Matching foundational ontologies			
Work	Ontologies	Approach	Available alignment
[27]	BFO 1.0, DOLCE	Manual comparison	–
[79]	BFO 1.0, DOLCE	Manual comparison	–
[85]	BFO 1.0, DOLCE	Manual alignment	Set of triples
[42]	BFO 1.1, DOLCE-Lite, GFO	Manual, matching tools	List at Romulus*
[78]	BFO 1.0, BFO 1.1,BFO 2.0	Semi-automatic (change-tracking)	–
[59]	SUMO, DOLCE	Manual alignment	–
[56]	SUMO, DOLCE-CORE	Manual alignment	FOL alignments
[73]	BFO, GFO, DOLCE-Lite, SUMO	Matching tools	Alignment API format
[40]	DOLCE-Lite, SUMO	Hypernym relation extraction	Alignment API format

Table 2 summarizes work on matching of foundational ontologies. They are mostly manual efforts, with a few of the resulting alignments made available.

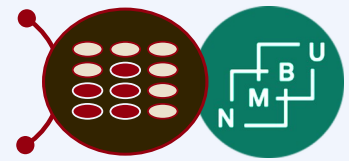
¹C. Trojahn, R. Vieira, et al., «Foundational ontologies meet ontology matching: A survey», *Semantic Web* **13**: 685–704, doi:10.3233/sw-210447, 2022

Concept schemes using SKOS

particular:	object entity individual	relationship	property (sometimes: attribute)
universal:	class entity type concept	relation relationship type (in OWL: ObjectProperty)	attribute (sometimes: attribute type) (in OWL: DatatypeProperty)

Definition: A **concept** is a universal that is only instantiated by individuals.

- From **SKOS**, a semantic artefact for organizing concept schemes: “Concepts are the units of thought - ideas, meanings, or (categories of) objects and events - which underlie many knowledge organization systems” (Isaac & Summers 2009).
- In many settings, including in object-oriented programming, a concept is usually called a **class**. In E-R diagram terminology, it is called an **entity type**.
- E-R terminology distinguishes between an entity type and the corresponding **entity set**, i.e., the set of all individuals that instantiate the entity type. In nominalist ontology, these two are the same – a universal is the set of its individual instances.



Concept schemes using SKOS

SKOS for taxonomies

Look at the exact meanings of **skos:broader** and **skos:narrower**.

SKOS for alignments

Imagine you have a knowledge graph in ontology C (namespace prefix C).

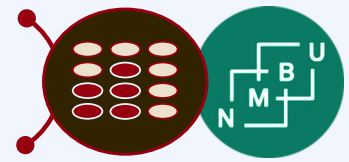
It is your task to transform it to ontology B (namespace prefix B). The target graph (knowledge graph obtained as solution) should only use B, no longer C.

The following are given in a SKOS alignment between B and C. Explain:

C:book **skos:exactMatch** **B:BookArtefact**.

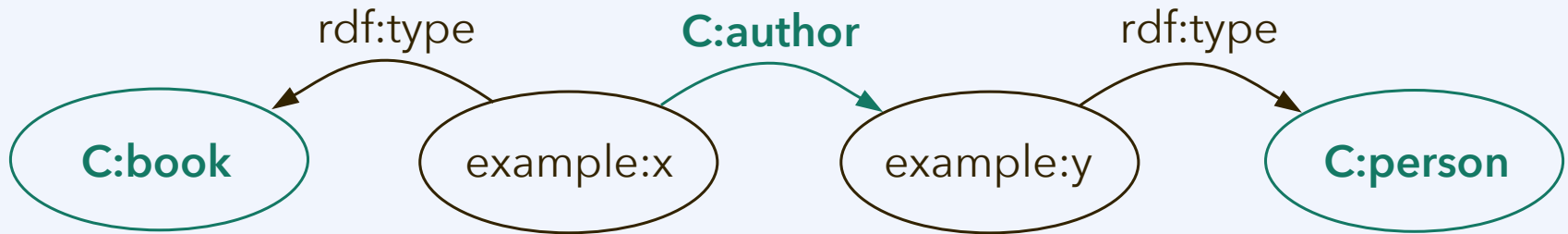
C:author **skos:broadMatch** **B:hasCreator**.

C:person **skos:narrowMatch** **B:Author**.



Short exercise: Mapping using SKOS

Imagine you have a knowledge graph in ontology C (namespace prefix C).



C:book skos:exactMatch B:BookArtefact.

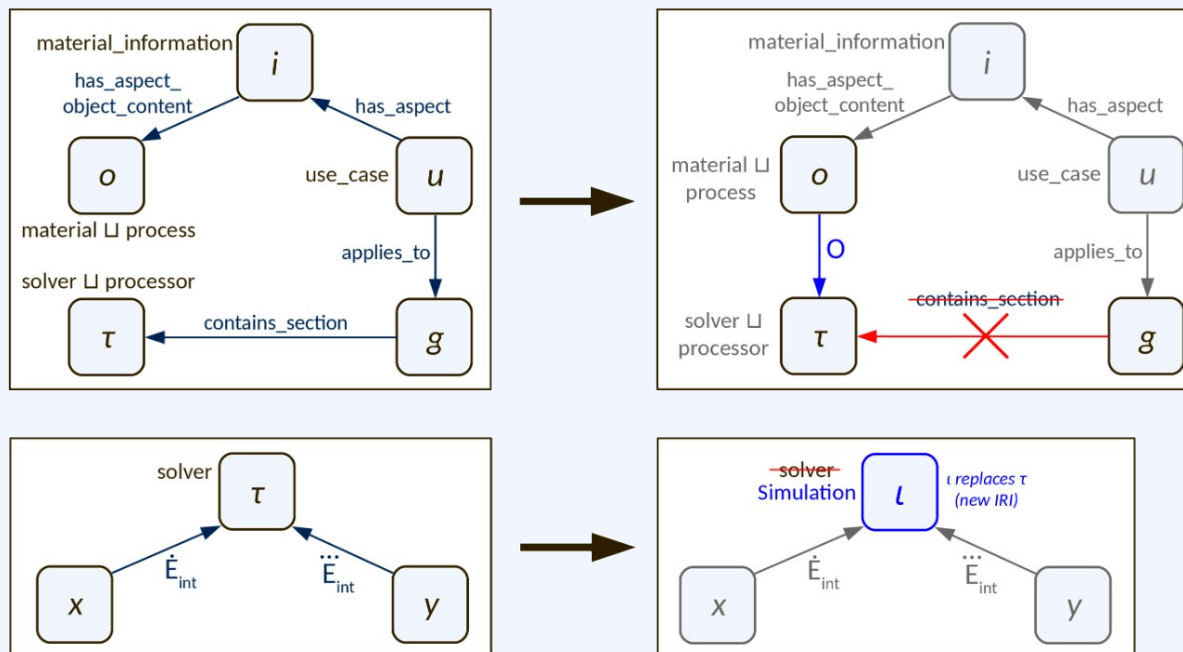
C:author skos:broadMatch B:hasCreator.

C:person skos:narrowMatch B:Author.

What graph do we obtain when mapping the one above to ontology B?

Semantic heterogeneity // graph transformation

Different platforms and tools use different metadata standards (ontologies). They are often so different that the shape of the data is different, not just terms.



Some 1:1 mappings between terms can be done using SKOS.

But we can retain more information if we can change the structure.

This can be done by applying **transformation rules** to the knowledge graph.¹

We can use **SPARQL** to formulate transformation rules.

¹P. Klein et al., in *Proceedings of JOWO 2021*, 2021.

Semantic heterogeneity // graph transformation

STTL: SPARQL Template Transformation Language¹

SPARQL is also provided with a CONSTRUCT WHERE query form the result of which is a graph. The CONSTRUCT clause specifies a graph pattern with variables which are replaced by the values found in the solutions of the WHERE clause in order to create a graph. For instance, the following SPARQL query enables to construct the RDF graph of foaf:knows inverse relations between the resources of the original RDF graph.

```
PREFIX foaf: <http://xmlns.com/foaf/0.1/>
CONSTRUCT {
  ?y foaf:knows ?x }
WHERE {
  ?x foaf:knows ?y }
```

This can be done by applying **transformation rules** to the knowledge graph.

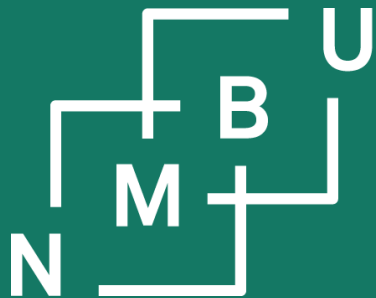
We can use **SPARQL** to formulate transformation rules.

3.2 SPARQL Template Query Form

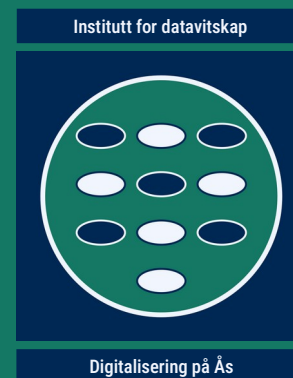
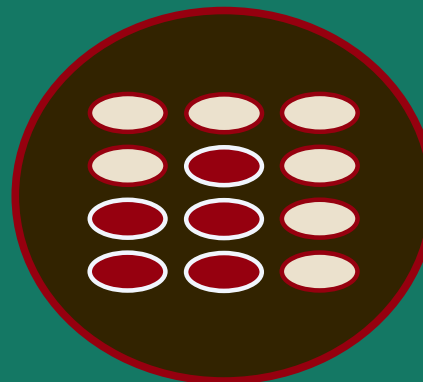
STTL relies on two extensions of SPARQL: an additional TEMPLATE query form and extension functions such as st:apply-templates to process a set of templates.

A TEMPLATE query is made of a standard WHERE clause and a TEMPLATE clause. The WHERE clause is the condition part of a rule, specifying the nodes in the RDF graph to be selected for the transformation. The TEMPLATE clause is the presentation part of the rule, specifying the output of the transformation for the RDF statements matching the condition.

¹O. Corby, C.F. Zucker, in *Proceedings of WEBIST 2015*: 318–340, **2016**.



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Datahandtering og -analyse

1 Ontologiar

- 1.1 Den semantiske veven
- 1.2 Vevontologispråket OWL
- 1.3 Terminologiutvikling

- 1.4 Ontologiutvikling
- 1.5 **BFO, DOLCE og EMMO**
- 1.6 **Semantisk heterogenitet**